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Study of begomovirus prevalently infecting tomato in Bihar

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Abstract- Over the past few decades, begomoviruses have steadily expanded across India, infecting a growing diversity of crop species and posing serious risks to agricultural productivity. In this study, we examined the diversity of begomoviruses affecting tomato crops in major tomato-growing districts of Bihar. Infected tomato plant field samples were collected during 2022-2023 across districts of Bihar, and total genomic DNA was extracted to screen for the presence of begomovirus DNA-A components. The amplified viral DNA-A and DNA-β fragments were cloned into the pJET1.2 vector, verified through PCR and restriction enzyme analysis, and subsequently identified by Sanger sequencing. Out of 10 monomeric DNA-A clones of begomovirus, 9 were identified as isolates of Tomato leaf curl virus and Tomato leaf curl New Delhi virus, along with 1 isolate of Papaya leaf crumple virus. Our findings reveal a concerning rise in incidences of Tomato leaf curl New Delhi virus infections in tomatoes along with diverse betasatellites across fields of Bihar.

Keywords: Biodiversity, Evolution of begomovirus, Plant disease in Gangetic plain, Tomato leaf curl.

INTRODUCTION

Tomato (*Solanum lycopersicum* L.) is one of the most economically and nutritionally significant vegetable crops cultivated in India. Our country ranks as the second largest producer of tomato globally (FAO 2019).[1] Despite this national prominence, Bihar contributes only about 5–6% of India's total tomato production. According to ICAR, approximately 46,000 ha of land in Bihar is under tomato cultivation. For marginal and smallholder farmers, tomato represents not only a dietary staple rich in vitamins, minerals, and antioxidants, but also a vital source of livelihood. However, the average productivity in Bihar (≈16 tonnes/ha) remains substantially lower than the national potential. Among several biotic constraints, viral diseases particularly those caused by begomoviruses, constitute a major limiting factor in tomato productivity.

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Begomoviruses (Genus- *Begomovirus*, Family- Geminiviridae) are among the most destructive pathogens of tomato and many other dicotyledonous hosts. Tomato Leaf Curl Disease (ToLCD), primarily caused by begomoviruses, is characterized by severe leaf curling, chlorosis, stunted growth, and reduced fruit set, frequently resulting in yield losses ranging from 30% to 100% under high vector pressure and favourable environmental conditions. [2-5] The economic consequences of such losses extend beyond individual farmers to regional vegetable supply chains and food security.

Tomato Leaf Curl Viruses (ToLCVs) possess circular single-stranded DNA genomes that may be monopartite (DNA-A) or bipartite (DNA-A and DNA-B). Their evolutionary dynamics are driven by mutation, recombination, genetic drift, and natural selection, enabling rapid adaptation to host defense mechanisms and environmental changes.[6,7] Over the past decade, both

monopartite and bipartite ToLCV species have been reported in nature, all transmitted by the whitefly *Bemisia tabaci*, a highly adaptable vector whose population dynamics are strongly influenced by climatic conditions. Although resistant cultivars are considered an economical management strategy, most available varieties exhibit only partial resistance against diverse ToLCV species. In Bihar and neighbouring Uttar Pradesh, multiple ToLCV species coexist, and early-stage infection can reduce yields by up to 80–90%.

Among bipartite Begomoviruses, Tomato Leaf Curl New Delhi Virus (ToLCNDV) was first reported in 1995 as a pathogen of tomato in northern India.[8] The virus comprises two circular single-stranded DNA components (~2.7 kb each), designated DNA-A and DNA-B, and is transmitted by *Bemisia tabaci*. ToLCNDV-encoded proteins are multifunctional and facilitate suppression of host defense responses, contributing to severe pathogenicity. Initially reported from northern India, the virus has expanded its geographical distribution to the Middle East, Mediterranean basin, and parts of Europe, reflecting its remarkable epidemiological adaptability.[9,10] According to the DBT (2020) risk classification of microorganisms, Tomato leaf curl New Delhi virus, Tomato leaf curl Bangalore virus, and Tobacco mosaic virus are placed in Risk Group 4, whereas Tomato leaf curl Patna virus and Tomato leaf curl Gujarat virus are categorized under comparatively lower risk groups (DBT list 2020).

In recent years, the incidence of ToLCVs in Bihar has increased markedly, with frequent outbreaks causing substantial yield reductions. Disease spread is influenced by warm and humid climatic conditions conducive to the proliferation of *Bemisia tabaci,* movement of infected planting material, inadequate vector management, and limited awareness of phytosanitary measures. The genetic variability and adaptive potential of begomoviruses further complicate disease management. Ongoing research efforts aim to elucidate molecular determinants of pathogenicity and virus–host interactions to support sustainable disease control strategies.

Additionally, the movement of infected plant materials, poor farm hygiene practices, and inadequate pest management strategies contribute to the virus's dissemination across different regions of Bihar. This study aims to identify the begomovirus molecules responsible for pathogenicity in tomato because of their counteracting nature with host defence mechanism and their tendencies for diversified adaptations capabilities. This study aims to recognise the begomovirus molecule responsible for pathogenicity among tomato plant and their interactive counterparts in the host.

MATERIALS & METHODS

Survey of ToLCV in different vegetable growing regions of Bihar-

Infected tomato samples showing symptoms like begomovirus infection (leaf curling, crumpling of leaf surface, vein clearing, mosaic pattern on leaf, leaf yellowing and upward or downward rolling of leaf margins) were collected during field survey (2022-2023) of vegetable growing regions of Bihar. The infected plant sample pictures were taken and symptoms were recorded (Fig.1 and Fig. 2). The collected samples were stored at -80°C for further research. The locations selected for the field study was based on the information collected from agricultural co-ordinators as well as from local people, through query about incidences of leaf curl, followed by visiting various fields across all districts of Bihar. Sample collection and survey location are marked with photograph of fields tagged with GPS locations. (Fig. 1)

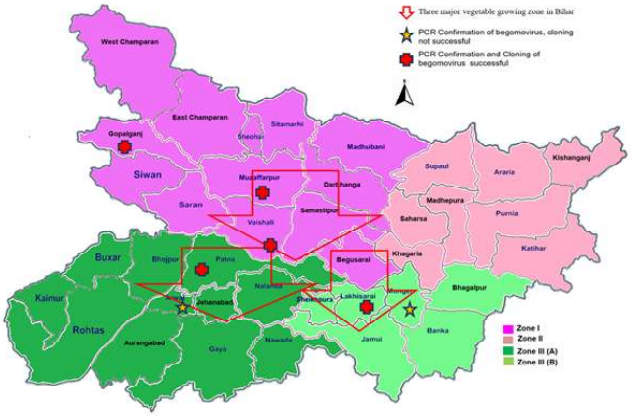


Fig. 1- Map of Bihar showing the three major vegetable growing zones visited in this survey, survey location of symptomatic tomato plant sample collection that tested positive for Begomovirus PCR and samples from where Begomovirus clones are obtained.

DNA isolation and cloning –

Total genomic DNA was extracted from plant samples using the CTAB method (modified from Cold Spring harbor). Begomovirus specific Roja's primers were used to confirm the presence of virus in collected sample.[11]

Confirmation of begomovirus presence was followed by PCR using Pfu polymerase to amplify viral DNA for cloning of full length DNA-A and DNA-β. For this, Kumar, (2020)[12], DNA-A primer and Briddon, (2002)[13], DNA-β primer are used to amplify full-length DNA-A (2.7 kb) and DNA-β (1.35 kb) genome respectively (Fig. 3), following the parameters as mentioned in the paper. Positively amplified genomes were cloned through blunt end ligation using pJET vector (Fermentas), as per the manual provided with the kit.



Sample Name- NPR1
Place- Distt.-Muzaffarpur, Village-Nawada Urf Khangura khurd
Date of collection-15/01/2023
Cloned genome – DNA-A (NPR1 01A)



Sample Name- NPR2
Place- Distt.-Muzaffarpur, Village-Nawada Urf Khangura khurd
Date of collection-15/01/2023
Cloned genome – DNA-A (NPR2 02A and NPR2 03A) and DNA-β (NPR2 04B)



Sample Name- NPR3
Place- Distt.-Muzaffarpur, Village-Nawada Urf Khangura khurd
Date of collection-15/01/2023
Cloned genome – DNA-A (NPR3 05A) and DNA-β (NPR3 06B and NPR3 07B)



Sample Name- NPR4
Place- Distt.-Muzaffarpur, Village-Nawada Urf Khangura khurd
Date of collection-15/01/2023
Cloned genome – DNA-A (NPR4 08A) and DNA-β (NPR4 09B)



Sample Name- PBS1
Place- Distt.-Gopalganj, Village-Bhringichak
Date of collection-15/01/2023
Cloned genome – DNA-A (PBS1 10A)



Sample Name- PRD1
Place- Distt.-Vaishali, Village- Daulatpur
Date of collection-31/12/2022
Cloned genome – DNA-A (PRD1 11A) and DNA-β (PRD1 12B)



Sample Name- PRD2
Place- Distt.-Vaishali, Village- Daulatpur
Date of collection-31/12/2022
Cloned genome – DNA-A (PRD2 13A) and DNA-β (PRD1 14B)

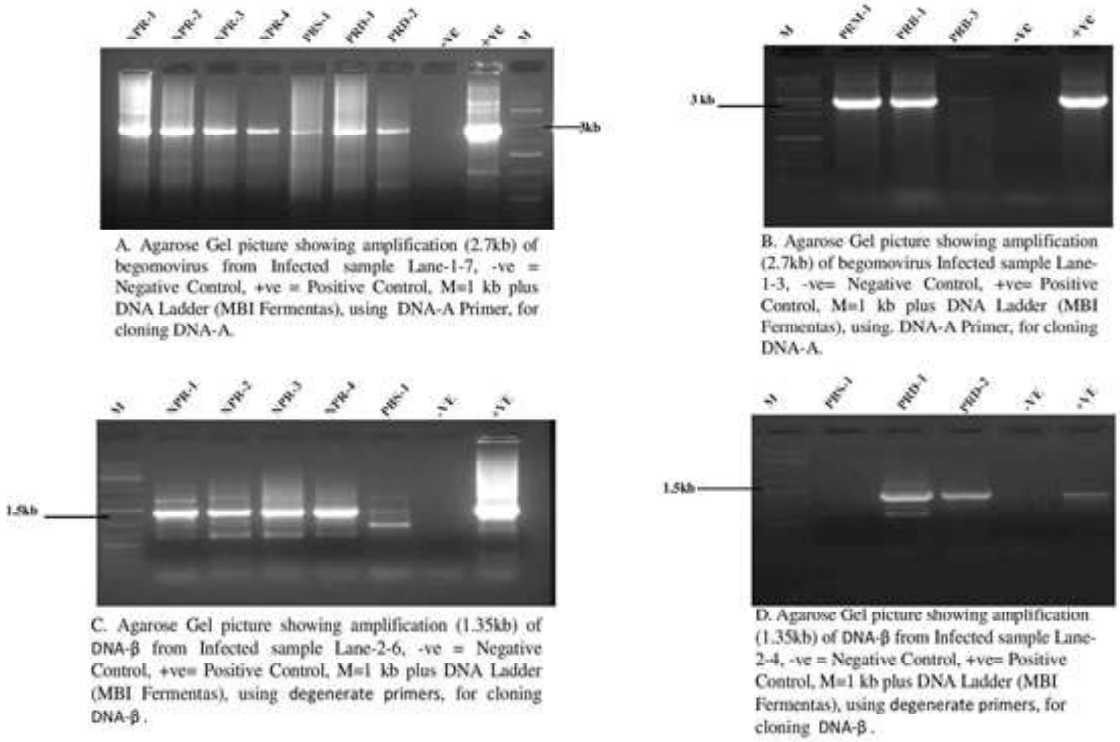


Sample Name- PRM1
Place- Distt.-Patna, Village-Kanhaipur
Date of collection-27/11/2022
Cloned genome – DNA-A (PRM1 15A)



Sample Name- PRB1
Place- Distt.-Lakhisarai, Village-Badhaiya
Date of collection-25/11/2022
Cloned genome – DNA-A (PRB1 16A)

Fig. 2- Pictures showing symptomatic plant in field condition, from where clones are constructed for DNA-A and DNA-β genome, date of collection, GPS location and sample name are mentioned in each picture.



A. Agarose Gel picture showing amplification (2.7kb) of begomovirus from Infected sample Lane-1-7, -ve = Negative Control, +ve = Positive Control, M=1 kb plus DNA Ladder (MBI Fermentas), using DNA-A Primer, for cloning DNA-A.

B. Agarose Gel picture showing amplification (2.7kb) of begomovirus Infected sample Lane-1-3, -ve= Negative Control, +ve= Positive Control, M=1 kb plus DNA Ladder (MBI Fermentas), using. DNA-A Primer, for cloning DNA-A.

C. Agarose Gel picture showing amplification (1.35kb) of DNA-β from Infected sample Lane-2-6, -ve = Negative Control, +ve= Positive Control, M=1 kb plus DNA Ladder (MBI Fermentas), using degenerate primers, for cloning DNA-β.

D. Agarose Gel picture showing amplification (1.35kb) of DNA-β from Infected sample Lane-2-4, -ve = Negative Control, +ve= Positive Control, M=1 kb plus DNA Ladder (MBI Fermentas), using degenerate primers, for cloning DNA-β.

Fig. 3- Agarose gel picture showing amplification of DNA-A and DNA-β, used for cloning in pJET 1.2 vector.

Sequencing of DNA-A and DNA-β clones and sequence analysis-

All the PCR positive samples of DNA were attempted for cloning, out of which the positive clones of DNA-A and DNA- β, were sent for partial sequencing to know the begomovirus existing in these regions. Total 10 (ten) DNA-A and 6 (six) DNA-β positive clones were sent for partial sequencing, at UDSC, CIIDRET, DU. The results were analysed through BLAST (blantn;megablast). The percentage similarity results and species identification on blastn basis are demonstrated in the Table 1.

RESULTS & DISCUSSION

A survey of tomato fields was conducted to assess the current distribution of begomoviruses, performed at selected vegetable growing regions across districts of Bihar, during November, 2022 to April, 2023, following outcomes are obtained.

Study of Tomato infecting begomovirus disease prevalence

Bihar has three major vegetable growing zones in Lakhisarai district, Nalanda district and at the Samastipur-Darbhanga-Madhubani Districts. Present survey has covered Lakhisarai (Village- Badhaiya) and Nalanda (Village-Shahoker-Sosarai) district. In Lakhisarai, incidence of Tomato leaf curl is recorded where tomato was growing on large scale but in Nalanda tomato cultivation was found on marginal ratio with almost no disease incidence. In addition to this vegetable growing zone on map of Bihar state is (Fig. 1) marked with places, covered under survey, identified with begomovirus infection and from where viral genome was successfully cloned (Fig. 2). GPS map picture of all visited places/fields with infected tomato plants are generated for future reference.

Assessment of DNA-A and DNA-β associated with tomato-infecting Begomoviruses

PCR amplification to confirm the presence of Begomovirus and cloning of replicative form of Viral DNA of the begomoviruses infecting tomato from selected sample successfully obtained as mentioned in methods. DNA-A primer and DNA-β primer are successfully used to amplify full-length DNA-A and DNA-β genome respectively (Fig. 3).[12,13] The amplicons cloned in pJET vector (Fermentas). Table 1 indicates the partial genome sequencing details of (10+6= 16 clones) and their BLAST (blastn; megablast) identification analysis results through NCBI (Table 1). Analysis of partial genome sequence data revealed that Tomato leaf curl virus (ToLCV) isolates and Tomato leaf curl New Delhi virus (ToLCNDV) isolates are

the most prevalently existing genome across the fields. Radish leaf curl betasatellites is found in association with ToLCNDV and Papaya leaf crumple virus is found in association with Croton yellow vein mosaic betasatellite, infecting tomato at Muzaffarpur districts. Two kinds of betasatellite molecule namely (as per partial sequencing data) Radish leaf curl betasatellites and Tomato leaf curl betasatellite are associated with ToLCNDV in Muzzafarpur and Okra enation leaf curl betasatellite is associated with ToLCNDV in Vaishali. From Vaishali, another sample infected with ToLCNDV associated with Radish leaf curl betasatellites isolate.

Table 1: Details of clones (16) for which partial sequencing data is obtained: DNA-A (10 clones) and DNA-β (6 clones) and their BLASTn analysis report for identification of species/isolate

Sl. No.	Place of collection	Sample Name (Genome cloned)	No. of base pair in partial sequence data	Blastn nearest match (Accession no.)	Query coverage, Percent Identity n Blastn with nearest match	(Partial sequence alignment in bp) genomic region
1.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR1 01A (DNA-A)	747 bp	Tomato leaf curl New Delhi virus (MW399221)	96%, 95.5%	855-135 (cds 135-855) pre coat protein and coat protein region
2.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR2 02A (DNA-A)	774 bp	Tomato leaf curl virus (OM777194)	98%, 99.2%	(298-1058) pre coat protein and coat protein region
3.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR2 03A (DNA- A)	784 bp	Tomato leaf curl New Delhi virus (MW399221)	98%, 95%	855-78 (cds 78-855) pre coat protein and coat protein region
4.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR2 04B (DNA-β)	740 bp	Radish leaf curl betasatellite isolate Salem (JN663873)	98%, 100%	(1294-1357 and 1-663) SCR and ORFβC1
5.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR3 05A (DNA- A)	785 bp	Tomato leaf curl virus (OM777194)	97%, 99.2%	(298-1066) pre coat protein and coat protein region
6.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR3 06B (DNA-β)	788 bp	Tomato leaf curl betasatellite isolate (MT 385292)	98%, 92.9%	(1281-1346 and 1-710) SCR and ORFβC1
7.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR3 07B (DNA-β)	777 bp	Tomato leaf curl betasatellite isolate (MT 385292)	98%, 92.7%	(1281-1346 and 1-698) SCR and ORFβC1
8.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR4 08A DNA -A	718 bp	Papaya leaf crumple virus isolate Bhopal (KX302712)	97%, 97.9%	(2342-2733 and 1-306) Ac4 protein, CR and Pre coat protein
9.	Dist.- Muzaffarpur Village- Nawada Urf Khangura khurd	NPR4 09B (DNA-β)	779 bp	Croton Yellow vein mosaic betasatellite (EU557375)	98%, 92.4%	(537-1301) A-rich and SCR region
10.	Distt.-Gopalganj Village- Bhringichak	PBS1 10A (DNA-A)	771 bp	Tomato leaf curl New Delhi virus isolate (MW399221)	98%, 98.2%	855-99 (cds 99-855) pre coat protein and coat protein region
11.	Distt.-Vaishali Village- Daulatpur	PRD1 11A (DNA- A)	779bp	Tomato leaf curl New Delhi virus isolate (OM048966)	98%, 98.6%	(298-1063) pre coat protein and coat protein region
12.	Distt.-Vaishali Village- Daulatpur	PRD1 12B (DNA-β)	780 bp	Okra enation leaf curl betasatellite isolate (OQ683403)	97%, 98.3%	(509-1270) A-rich and SCR region
13.	Distt.-Vaishali Village- Daulatpur	PRD2 13A (DNA-A)	782 bp	Tomato leaf curl New Delhi virus isolate (OM048966)	97%, 98.9%	(298-1061) pre coat protein and coat protein region
14.	Distt.-Vaishali Village- Daulatpur	PRD2 14B (DNA-β)	734 bp	Radish leaf curl betasatellite isolate Salem (JN663873)	98%, 100%	(1-657 and 1294-1357) SCR and ORFβC1
15.	Distt.-Patna (Mokama) Village- Kanhaipur	PRM1 15A (DNA-A)	778 bp	Tomato leaf curl virus (OM777194)	98%, 98.7%	(298-1063) pre coat protein and coat protein region
16.	Distt.-Lakhisarai Village-Badhaiya	PRB1 16A (DNA-A)	781 bp	Tomato leaf curl virus (OM777194)	98%, 98.9%	(298-1064) pre coat protein and coat protein region

Begomoviruses associated with betasatellites are widely distributed across the Indian subcontinent, although many species exhibit relatively restricted geographical ranges. In contrast, ToLCNDV, an Old World bipartite begomovirus, demonstrates an exceptional epidemiological footprint extending beyond South Asia to regions of the Far East, Middle East, North Africa, and parts of Europe.[8,14] This remarkable dispersal capacity has been attributed to the high evolutionary plasticity of its DNA-A component. The proteins encoded by DNA-A are known to functionally interact with diverse DNA-B components and, in some cases, betasatellites, enabling transcomplementation and facilitating successful infection in varied host backgrounds.[7,15-17] Such interactions enhance opportunities for recombination and genomic reassortment, processes that contribute to host range expansion and viral adaptation.

In the present investigation, ToLCNDV was found to be the predominant begomovirus infecting tomato in Muzaffarpur, Vaishali, and Gopalganj districts of Bihar, frequently detected in association with distinct betasatellites. This pattern supports earlier observations that satellite molecules may influence symptom severity and pathogenic fitness.[18] The widespread occurrence of ToLCNDV in multiple agro-climatic zones of Bihar reflects both its adaptive potential and the conducive ecological conditions that favor vector-mediated transmission.

Tomato Leaf Curl Virus (ToLCV) emerged as the next most prevalent species in the present study, particularly in Patna and Badhaiya districts. Notably, in Muzaffarpur district, Papaya leaf crumple virus was detected infecting tomato in association with Croton yellow vein mosaic betasatellite, indicating successful cross-host establishment. Such findings underscore the dynamic host adaptability of begomoviruses and their capacity to establish novel host–pathogen interactions under field conditions.

Host range expansion among ToLCVs has been well documented. For instance, Tomato leaf curl Patna virus and Tomato leaf curl Palampur virus initially reported in tomato, have subsequently been identified in additional host species.[3,19] Furthermore, pseudo-recombination events between ToLCNDV and Tomato leaf curl Palampur virus have been observed both under natural field conditions and in experimental systems, highlighting the propensity of these viruses for genomic reassortment.[20] The concurrent processes of host adaptation and genome reconstitution appear to operate synergistically, enhancing viral fitness and epidemiological success.

CONCLUSION

Collectively, the findings of the present study reinforce the emerging prominence of ToLCNDV as a highly adaptable and evolutionarily dynamic begomovirus in Bihar. A deeper understanding of its molecular determinants of pathogenicity, satellite interactions, and recombination potential is essential for anticipating future outbreaks and mitigating economic losses. Monitoring ToLCNDV prevalence along with associated betasatellites enables early detection of emerging variants with altered host range or enhanced virulence. Such integrative studies are critical for designing durable resistance strategies and effective disease management programs in tomato-growing regions. Further these insights into the ecological and evolutionary drivers of ToLCNDV dissemination may also inform broader strategies for managing begomoviral invasions in diverse agro-ecosystems.

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CONFLICT OF INTEREST

Authors declare no conflict of interest.

REFERENCES

1. **Food and Agriculture Organization 2019.** FAOSTAT statistical database, FAO, Rome.
2. **Vasudeva R. S. & Sam Raj J. 1948.** A leaf-curl disease of tomato, *Phytopathology* 38(5).
3. **Kumar Y., Hallan V. & Zaidi A. A. 2008.** Molecular characterization of a distinct bipartite begomovirus species infecting tomato in India. *Virus Genes.* **37(3):** 425-431.
4. **Kumari P., Singh A. K., Chattopadhyay B. & Chakraborty S. 2010.** Molecular characterization of a new species of begomovirus and betasatellite causing leaf curl disease of tomato in India. *Virus Res.* **152(1-2):** 19-29.

5. **Sangeeta, Ranjan P., Kumar R. V., Bhatt B. S., Chahwala F. D., Yadav B. K., Patel S., Singh B. & Singh A. K. 2021.** Two distinct monopartite begomovirus-betasatellite complexes in western India cause tomato leaf curl disease, *Virus Res.* **198:** 319.

6. **Varma A. & Malathi V. G. 2003.** Emerging geminivirus problems: a serious threat to crop production, *Ann. Appl. Biol.* **142(2):** 145-164.

7. **Nawaz-ul-Rehman M. S. & Fauquet C. M. 2009.** Evolution of geminiviruses and their satellites, *FEBS Lett.* **583(12):** 1825-1832.

8. **Srivastava K. M., Hallan V., Raizada R. K., Chandra G., Singh B. P. & Sane P. V. 1995.** Molecular cloning of Indian tomato leaf curl virus genome following a simple method of concentrating the supercoiled replicative form of viral DNA, *J. Virol. Methods.* **51:** 297-304.

9. **Panno S., Iacono G., Davino S., Marchione S., Zappardo V., Bella P., Rangel E. A. & Davino M. 2016.** First report of Tomato leaf curl New Delhi virus in Italy, *Plant Dis*. **100:** 2335.

10. **Fortes I. M., Sánchez-Campos S., Fiallo-Olivé E., Díaz-Pendón J. A., Navas-Castillo J. & Moriones E. 2016.** A novel strain of Tomato leaf curl New Delhi virus has spread to the Mediterranean basin, Plant Pathol. 65: 64-75.

11. **Rojas M. R., Noueiry A. O., Lucas W. J. & Gilbertson R. L. 1998.** Bean dwarf mosaic geminivirus movement proteins recognize DNA in a form- and size-specific manner, *Cell.* **95(1):** 105-113.

12. **Kumar M., Vinoth Kumar R. & Chakraborty S. 2020.** Association of a begomovirus-satellite complex with yellow vein and leaf curl disease of hollyhock (Alcea rosea) in India, *Arch. Virol.* **165(9):** 2099-2103.

13. **Briddon R. W., Bull S. E., Mansoor S., Amin I. & Markham P. G. 2002.** Universal primers for the PCR mediated amplification of DNAβ: a molecule associated with some monopartite begomoviruses, *Mol. Biotechnol.* **20:** 315-318.

14. **Zaidi S. S. A., Martin D. P., Amin I., Farooq M. & Mansoor S. 2017.** Tomato leaf curl New Delhi virus: a widespread bipartite begomovirus in the territory of monopartite begomoviruses, *Mol. Plant Pathol.* **18(7):** 901-911.

15. **Briddon R. W. & Stanley J. 2006.** Subviral agents associated with plant single-stranded DNA viruses, *Virology*. **344:** 198-210.

16. **Prasanna H. C. & Rai M. 2007.** Detection and management of Tomato leaf curl virus, *J. Plant Dis. Prot.* **114:** 65-72.

17. **Zhou X. 2013.** Advances in understanding begomovirus-betasatellite interactions, *Annu. Rev. Phytopathol.* **51:** 357-381.

18. **Briddon R. W., Mansoor S., Bedford I. D., Pinner M. S., Saunders K. & Stanley J. 2001.** Satellite DNAs associated with monopartite begomoviruses, *Virology* **280:** 97-109.

19. **Kumari P., Singh A. K., Chattopadhyay B. & Chakraborty S. 2011.** A new begomovirus species causing tomato leaf curl disease in Ranchi, India, New Dis. Rep. **23:** 11.

20. **Malik A. H., Briddon R. W. & Mansoor S. 2011.** Infectious clones of Tomato leaf curl Palampur virus with a defective DNA B and their pseudo-recombination with Tomato leaf curl New Delhi virus, *Virol. J.* **8:** 173.

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